Section: **4.3 – CRACK INIITATION ANALYSIS**

Chapter: **4330**

Title: **ACOUSTIC FATIGUE**

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1 INTRODUCTION

This chapter presents the methodology to be considered for the assessment of acoustic fatigue, also called sonic fatigue. The complete procedure for the determination of the acoustic excitation is out of the scope of this document and it is supposed to be an input for the fatigue analysis.

The acoustic fatigue studies the fatigue damage caused in a structure due to very high-frequency low-amplitude stress fluctuations associated with random acoustic loading (Ref. 2). In the field of the acoustic fatigue the cyclic stresses are low in amplitude compared with normal fatigue standards, but high in frequency (100 – 1000 Hz). If the acoustic excitation frequencies match any natural frequency of the structure, the vibrations levels will increase and will lead to a stress variation in addition to any stress already present in the structure.

In service experience has demonstrated that acoustic loading can cause the failure of aircraft structures such as engine pylons and engine mounts, control surfaces, skin panels, fuselage stringers, ribs, tailplane structures, roll equipment and stores. These failures are not catastrophic but lead to unacceptable maintenance costs. In addition to the analysis of the structure, the systems attachments have to be verified also in order to demonstrate enough acoustic fatigue endurance (see Ref. 7).

Acoustic fatigue is, in essence, a random vibration fatigue problem. Due to the random nature of the vibrations in this type of analysis, the most efficient way to solve it is to use a statistical process to calculate the probability of occurrence of amplitude. The random vibration is characterized by the mean, the standard deviation and the probability distribution. The cumulative fatigue damage is then calculated for a time period as the average of the damage of a number of cycles, instead of calculating the damage of each individual amplitude vibration.

The common practice to avoid acoustic fatigue problems is to demonstrate a fatigue life, with the corresponding scatter factor, over the DSG of the aircraft. On this basis, it is assumed that the MRB document will cover the safety of the component. Otherwise, a design modification can be required to move the natural frequencies to achieve that the new acoustic damage is acceptable.

2 METHODOLOGY

The general method to assess the acoustic fatigue is based on a series of ESDU items.

The sound pressure level is supposed to be an input for the fatigue analysis, but it must be transformed into consistent units.

The second step of the method is to calculate the natural frequencies and the damping of all the surfaces directly exposed or connected to a surface exposed to an acoustic environment.

Once the natural frequencies have been calculated, the stress response of the structure to the sound pressure level is calculated and compared, taking into account the stress concentration factor, with the material properties for random loads to predict the fatigue life.

ESDU method can be applied to a wide range of typical structural configurations and has been accepted by the airworthiness authorities for use in showing compliance to the regulations covering acoustic fatigue. Only when the structural configuration does not allow the application of the ESDU items, the acoustic assessment is performed using Dynamic FE models and PSD analysis. The FE models used for the acoustic fatigue analyses are required to be very fine and need an accurate modelling of the boundary conditions and the mass distribution.

Currently it is not possible to reliably predict the rate of fatigue crack growth in the presence of acoustic excitation, since the structural response changes as the damage progresses, continually altering the constraints and deflections in the structure. The approach is therefore to demonstrate that the affected structure is free from acoustic fatigue damage within the DSG. This is, essentially, a “safe-life” methodology (Ref. 7).

The next flowchart shows the steps to be followed in the acoustic fatigue calculation. Each step is explained in next chapters.

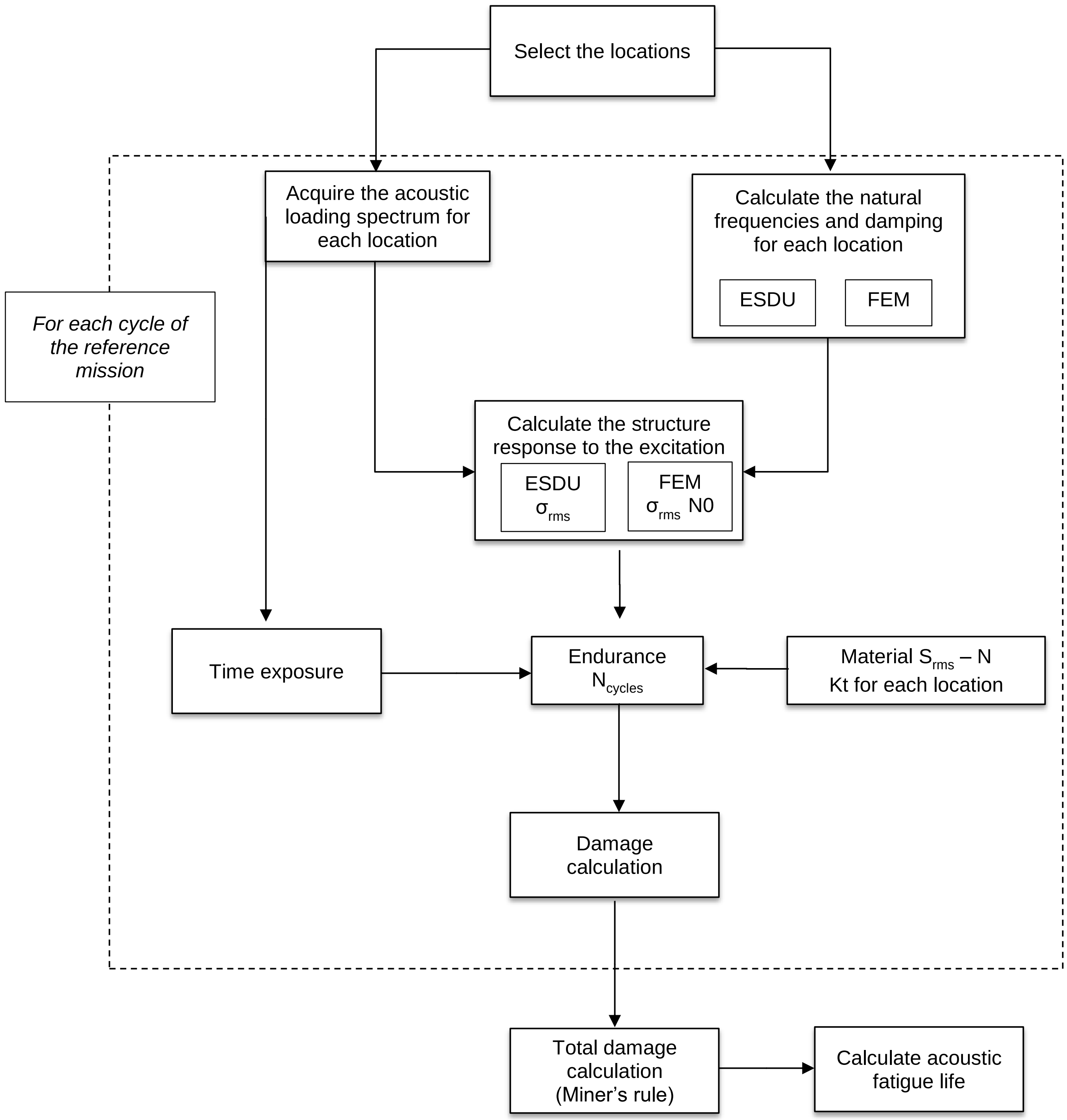

Figure 1. Acoustic fatigue life calculation

2.1 Affected areas

All the structural elements classified as category A and B and susceptible to acoustic fatigue must be considered, as well as the parts that are considered as having economic consequences of premature repair or replacement or are considered to compromise the airworthiness.

The list of structural parts susceptible to acoustic fatigue for an aircraft with a conventional configuration typically includes the following elements (Ref. 7):

- Wing box (spar and covers) and fixed leading edge.
- Engine pylons and engine mounts.
- Structure around APU (particularly the inlet structure).
- Control surfaces; inner and outer flaps, slats and other leading edge devices, spoilers and ailerons.
- Centre and rear fuselage, vertical fin, rudder, horizontal tailplane and elevator.
- Roll equipment, stores, bus of antennas.

The list of affected areas may vary from one program to another. The locations susceptible to acoustic damage must be identified at the start of the certification program and agreed with the airworthiness authorities.

2.2 Acoustic loading spectrum

The acoustic loading spectrum is obtained from each noise source and for each surface involved.

The spectrum is obtained from flight test data and it is always an input for the acoustic fatigue analysis. The acoustic spectrum is always expressed in terms of dB and should cover a frequency range that includes all the significant acoustic energy. It must clearly state the bandwidth used (usually in third-octave band[1]).

The most common methods used to characterize the sound spectra are the Sound Pressure Level (SPL) and the Power Spectral Density (PSD):

SOUND PRESSURE LEVEL (SPL)

The sound pressure (*p*), measured in Pascal (Pa), is defined as the difference between the pressure caused by a sound wave and the ambient pressure of the medium the sound wave is passing through. The most common way to measure the sound pressure is the root-mean-square (r.m.s.) pressure. The sound pressure level (SPL) is a logarithmic measure (in dB) of the effective sound pressure of a sound relative to a reference value:

[1] In an octave frequency band, the upper frequency is twice the lower frequency ($f_u = 2f_l$).
A third-octave band frequency is the third part of an octave and, i.e. each octave is divided into three frequency bands, being $f_u = 2^{1/3}f_l$. Since each of the octave bands is split into three, third-octave bands gives a more detailed description of the frequency content of the noise (Ref. 8).

$$\mathrm{SPL} = 20 \cdot \log\left[\left(\frac{p_{rms}}{p_{ref}}\right)\right]$$

Where p_{rms} is the effective root mean square sound pressure and p_{ref} refers to the sound pressure in air, (p_{ref} = 20µPa r.m.s). It corresponds to a sound pressure level of 0 dB.

Figure 2 shows an example of in-flight measurements, where third-octave band sound pressure level (SPL) in dB for a flap support fairing is provided:

FSF1	1/3 Octave Band Centre Frequencies (Hz)											
Segment	50	63	80	100	125	160	200	250	315	400	500	630
Take-Off	135,6	140,4	138,5	138,5	138,7	134,4	135,5	134,4	133,4	132,1	130,4	128,9
Max.Climb	119,6	130,0	123,5	137,3	129,4	127,7	142,0	129,1	134,8	132,8	134,8	132,6
Climb	119,9	119,2	125,4	143,1	126,1	126,9	138,8	130,7	136,8	136,1	133,5	131,3
Climb	112,9	109,5	125,0	135,6	114,9	116,6	132,5	121,9	130,7	129,5	129,4	129,6
Climb	119,7	123,1	122,6	142,3	127,2	125,8	140,7	127,4	131,3	133,2	131,4	130,9
Cruise	119,7	123,1	122,6	142,3	127,2	125,8	140,7	127,4	131,3	133,2	131,4	130,9
LNC	114,5	119,3	127,3	137,1	118,4	125,9	135,7	132,9	128,0	131,8	132,2	128,8
LLF	121,2	122,6	142,5	139,2	125,3	136,1	133,5	134,6	132,2	132,4	131,4	129,8
Approach	117,7	122,2	126,3	130,4	120,8	123,7	124,2	122,8	122,5	123,3	123,6	123,5

Figure 2. In-flight measurements (from Ref. 9)

The representation in third-octave band implies a variable frequency bandwidth, which is not suitable for the application of the ESDU methodology, which is based on constant frequency bandwidths of 1Hz at the natural frequency of the panel. The sound pressure level at a particular frequency and the sound pressure level in third-octave band centred at that frequency (f_c) are related through the following expression:

$$SPL = SPL_{1/3} - 10 \cdot \log f + 6.3533$$

POWER SPECTRAL DENSITY (PSD)

For complex structures for which the ESDU methodology is not applicable, the use of dynamic FE models is required. In this case, the acoustic spectrum must be defined through its Power Spectral Density.

The PSD can be obtained from the SPL by squaring the pressure levels and dividing by the octave band frequency range or third-octave:

$$PSD = \frac{\left(P_{ref} \cdot 10^{SPL/20}\right)^2}{\Delta f}$$

Where $p_{ref} = 2 \cdot 10^{-5}$ Pa and Δf is the bandwidth of the band.

TIME EXPOSURES

The structure is subjected to acoustic loading of different intensities, with noise exposure times that vary according to the fatigue mission flight plan. Note that the acoustic fatigue mission may be different from the fatigue missions because the acoustic loading can be different to the duration of the corresponding flight stage and the critical conditions are different. Therefore, an estimation of the acoustic fatigue mission may be preferable and should be included in the calculations.

2.3 Structural response

2.3.1 *Natural frequencies*

The acoustic fatigue analysis implies the analysis of the response of the structure to acoustic excitation. Therefore, knowledge of the modes of vibration of the structure and the natural frequencies and damping is required for the analysis:

- For simple geometries, ESDU items may be used to calculate the natural frequencies and damping:
  - Natural frequencies: Table 4.1, Table 4.2 and Table 4.3 of ESDU 74037 (Ref. 2) can be used as source of data for isotropic planes, orthotropic or laminated plates and sandwich panels and shell and box structures respectively.
  - Damping: ESDU 73011 (Ref. 4) provides the damping values for typical structures. ESDU 85012 (Ref. 5) can be used for the estimation of the damping factor of fibre-reinforced composite laminated plates.
- For complex geometries, the normal modes of the structural part are obtained with FE models by a modal analysis (see Ref. 8).

2.3.2 *Stress response to acoustic loading*

Once the structural damping ratio and the SPL at the resonant frequency are known, the root means square (r.m.s) stress at each natural frequency is calculated as a function of the spectral pressure level, the damping coefficient and the maximum stress form a unit pressure load.

The ESDU method is based on a simplified theory for stress response, which assumes that the dominant vibration of the structure is in its fundamental mode only. It is assumed also that the spectrum level of acoustic pressures is constant over the frequency range close to that fundamental frequency. In these conditions, the r.m.s stress of an isotropic panel at the probable failure location is defined as:

$$\sigma_{rms} = \left[\frac{\pi}{4\delta} f_n\right]^{1/2} \cdot SPL(f_n) \cdot \sigma_0$$

Where:

- f_n is the frequency of the fundamental mode.
- δ is the viscous damping ratio of the vibration mode at frequency f_n.
- $SPL(f_n)$ is the spectrum level of acoustic pressure at natural frequency of the panel (f_n).

  If there is more than one noise source, the levels from all of them have to be added following the technique given in ESDU 17012 (Ref. 6):

$$SPL_{sum} = 10 \cdot log\left[10^{SPL_1/10} + 10^{SPL_2/10} + \cdots + 10^{SPL_n/10}\right]$$

- σ_0 is the maximum static stress due to a unit uniform pressure acting over the critical damage location. It can be calculated using an open source method.

For laminated panel, the r.m.s strains instead of r.m.s. stresses have to be calculated because the stress would vary from layer to layer.

Basic values of σ_{rms} can be obtained from ESDU Items (see Table 1).

TYPE OF STRUCTURE	DATA SOURCE	REMARKS
Flat or curved isotropic skin-stringer panels having stiffeners of relatively high flexural stiffness (isotropic)	ESDU Item No. 72005	Applicable to stiffened panels where the flexural stiffnesses of the stiffeners are sufficiently high for a fixed-edge condition to be approximated for the individual plates
Flat or curved laminated skin-stringer panels having stiffeners of relatively high flexural stiffness	ESDU Item No. 84008	Applicable to stiffened panels where the flexural stiffnesses of the stiffeners are sufficiently high for a fixed-edge condition to be approximated for the individual plates
Flat or curved isotropic skin-stringer panels having stiffeners with flexural and torsional stiffnesses of the same order of magnitude	ESDU Item No. 73014	The condition where stiffener flexural and torsional stiffnesses are of the same order of magnitude is commonly found in integrally-machined panels
Flat or shallow curved sandwich panels having isotropic face plates of equal thickness	ESDU Item No. 72017	
Sandwich panels having laminated face plates of unequal thickness	ESDU Item No. 86024	Particularly applicable to sandwich panels having fibre-reinforced laminated face plates
attachment line of a solid edge member and the ratio of the surface strains on the inner and outer face plates at the centre of the panel	ESDU Item No. 95026	
Box structure internal plates and flanges	ESDU Item No. 74026	This type of structure is commonly found in aircraft tailplane and control-surface structures which, on jet aircraft, may lie close to the jet efflux, a region of high intensity acoustic loading.

Table 1. Data sources for the calculation of r.m.s. stresses or strains

For those complex structures for which ESDU methodology is not suitable, the structure response to the excitation can be obtained from dynamic FE models. The results will be:

- σ_{rms}
- N_0: It is the typical frequency of σ_{rms}, defined as the average number of zero crossings with positive slope per unit time.

The complete procedure is explained in Ref. 8.

2.4 Material fatigue properties under random vibration

In order to predict the acoustic fatigue life of a structural component, it is necessary to have random stress-life data (S_{rms} – N curve). The S_{rms} – N curves can be obtained either from dedicated test coupons excited by a random loading or derived from the standard constant amplitude coupon test assuming that the random signal distribution follows a Rayleigh distribution with appropriate factors to convert from sinusoidal to random excitation.

The S_{rms} – N data gives the life to failure for a material subjected to random vibration at specific root mean squared value of stress.

Table 2 shows the applicable ESDU Items for the calculation of data on the endurance of various types of structure subjected to acoustic loading. It is important to ensure that the fatigue strength data used are applicable to the detailed structural configurations because the highest stresses usually occur in the region of stiffener or frame rivet lines where the effects of stress concentration and fretting are important.

MATERIAL-STRUCTURAL DETAIL	DATA SOURCE
Joints in aluminium alloys	ESDU Item No. 72015
Joints in titanium alloys	ESDU Item No. 73010
Fibre-reinforced composite structural elements	ESDU Item No. 84027

Table 2. Data sources for the endurance under acoustic loading

It is important to note that stress concentrations factor due to loads and geometry must be considered when determining the value and location of the maximum stress.

2.5 Fatigue life calculation

Finally, the acoustic fatigue endurance is calculated as follows:

- The allowable number of cycles (N_{cyc}) for each segment of the flight profile is derived from S_{rms} – N curve using the σ_{rms} calculated from either the dynamic FE model or ESDU items (see chapter 2.3.2).

  The endurance in terms of flight hours is calculated as: $FH_i = N_{cyc,i}/(f/3600)$ where f is the excitation frequency.

- The damage for each segment is then calculated as the time of exposure of the segment divided by the endurance in terms of flight hours:

$$d_i = \frac{(Time\ of\ exposure)_i}{FH_i}$$

- The total damage for the DSG is calculated by applying the cumulative damage Miner's rule.

$$D = \sum d_i = \sum \frac{(Time\ of\ exposure)_i}{FH_i}$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
Airbus DS	---	Airbus Defence and Space
BPF	Hz	Blade Pass Frequency
D,d		Damage
dB		Decibels
DSG		Design Service Goal
ESDU		Engineering Sciences Data Unit
f	Hz	frequency
FH		Flight Hours
Hz		Hertz
N,n		Cycles
No		Number of zero crosses with positive slope
PSD		Power Spectral Density
rms		Root mean square
SPL	dB	Sound Pressure Level
δ		Damping
σ_{rms}		Stress rms

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	ESDU 86025	--	Design against fatigue: vibration of structures under acoustic or aerodynamic excitation
Ref. 2	ESDU 74037	--	Introduction and guide to ESDU data on acoustic fatigue
Ref. 3	ESDU 67028	---	Estimation of the RMS stress in skin panels subjected to random acoustic loading
Ref. 4	ESDU 73011	---	Damping in acoustically excited structures
Ref. 5	ESDU 85012	---	Estimation of damping in laminated and fibre-reinforced plates
Ref. 6	ESDU 17012	---	Combination of levels in dB
Ref. 7	ME-FA-NT-120123	--	Criteria for the analysis of sonic fatigue
Ref. 8	TEAG-IF-1910001	2019	Acoustic Fatigue. Task description & Calculation Philosophy

Ref. 9 ME-A4-DT-110158 A A400M Parts under AM responsibility Acoustic Fatigue

SOFTWARE

Program	***Version***	***Description***

Figure 3. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 07/11/2022	M.Carmen Blanquer Darío Fernández Javier Gómez-Escalonilla	First issue	All pages

Figure 4. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Analysis (TEASA)	Javier Gómez-Escalonilla 07/11/2022	Holger Hickethier 07/11/2022
Stress Methods (TEASP-TL3)	Darío R. Fernández Villalba 07/11/2022	-

Figure 5. Approval signatures table for last revision of this chapter